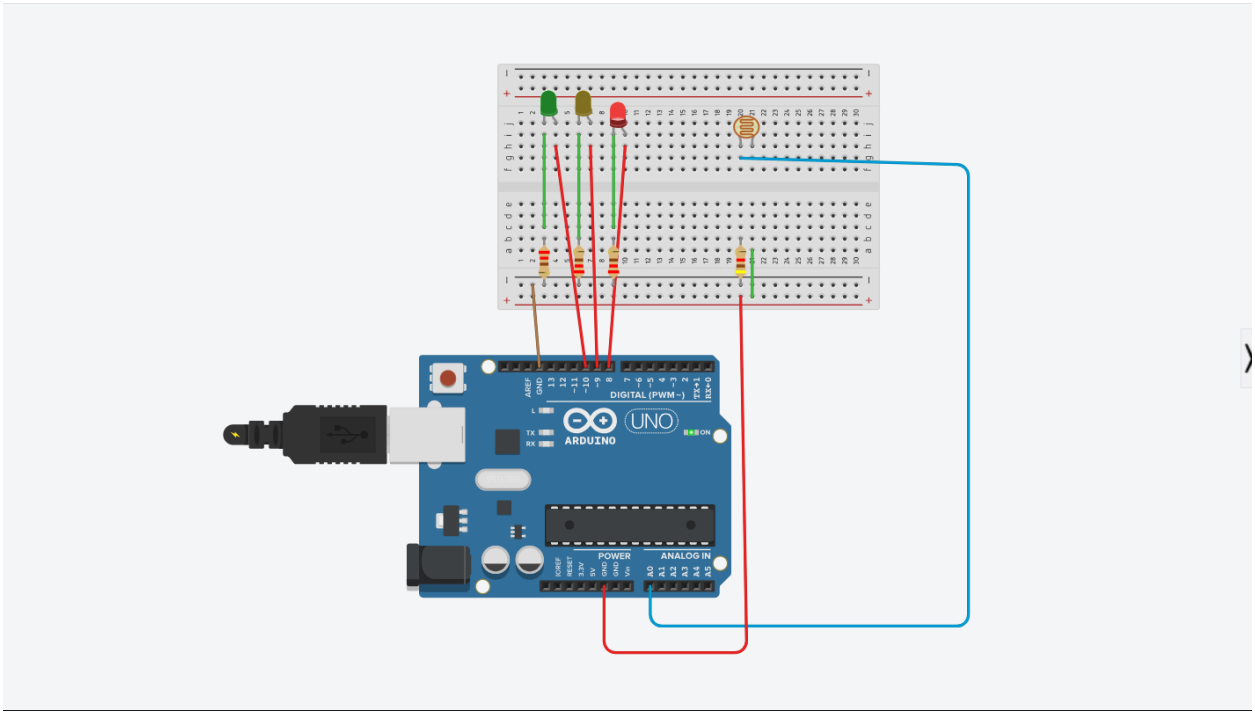
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Submitted to: Prof. Muddassar Sindhu

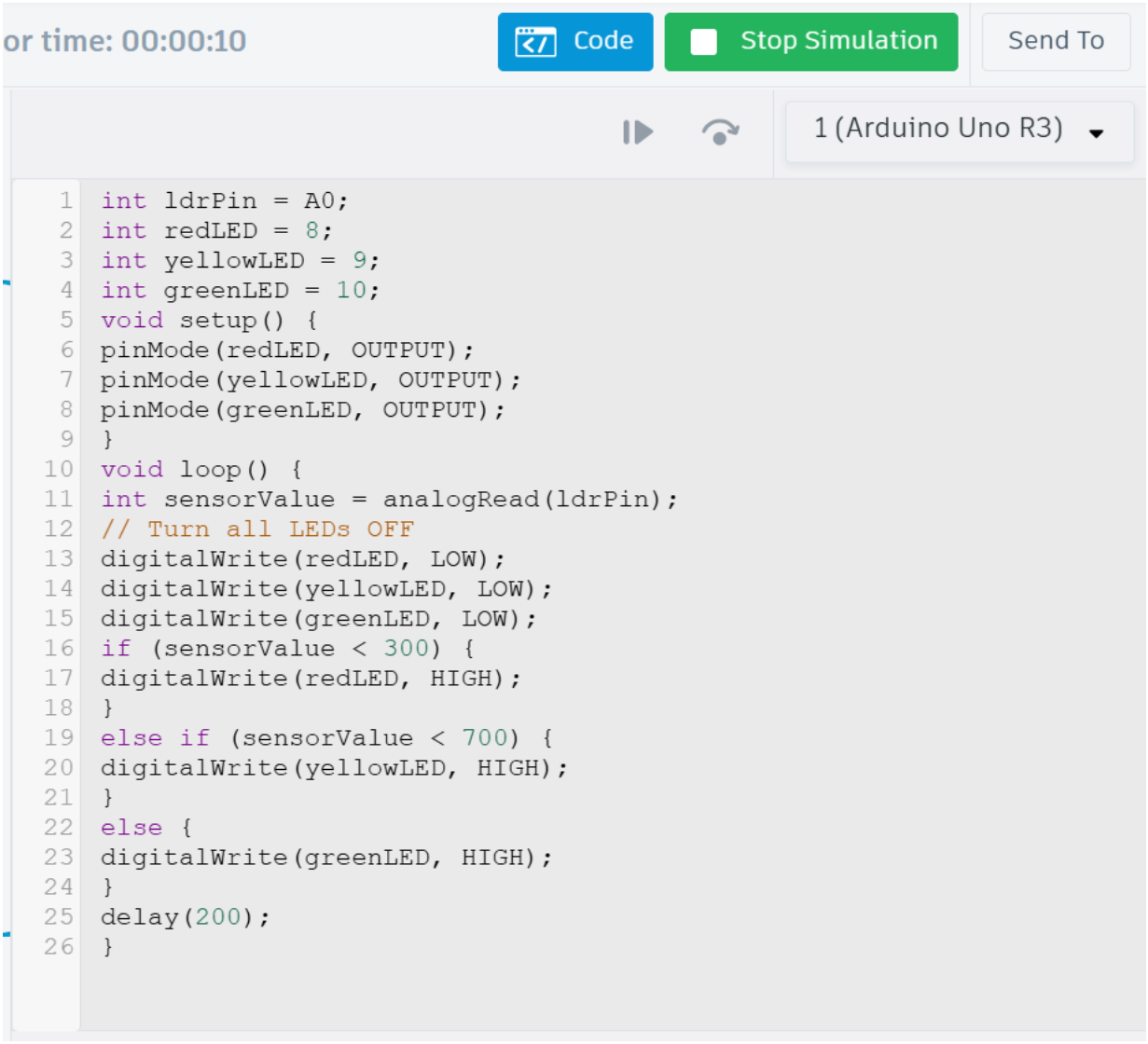
IoT Lab Task: Light Intensity Indicator Using LDR + Arduino UNO

Course: Programming for Data Science

Part 1: Circuit Setup



Part 2: Arduino Code




```
int ldrPin = A0;
int redLED = 8;
int yellowLED = 9;
int greenLED = 10;
void setup() {
pinMode(redLED, OUTPUT);
pinMode(yellowLED, OUTPUT);
pinMode(greenLED, OUTPUT);
}
void loop() {
int sensorValue = analogRead(ldrPin);
// Turn all LEDs OFF
digitalWrite(redLED, LOW);
digitalWrite(yellowLED, LOW);
digitalWrite(greenLED, LOW);
if (sensorValue < 300) {
digitalWrite(redLED, HIGH);
}
else if (sensorValue < 700) {
digitalWrite(yellowLED, HIGH);
}
else {
digitalWrite(greenLED, HIGH);
}
delay(200);
}
```

Part 4: Reflection

- The LDR (Light Dependent Resistor) changes its resistance based on the amount of light—low resistance in bright light and high resistance in darkness.
- When the LDR is paired with a 10kΩ resistor in a voltage divider, the Arduino can read the changing voltage as an analog input.
- Analog input gives a continuous value between 0 and 1023 instead of a simple ON/OFF reading, allowing detection of different light intensities.
- Threshold values (300 and 700) were used to divide sensor readings into three levels: dark, medium light, and bright light.

- During testing, the analogRead() values were observed in the Serial Monitor, and the thresholds were adjusted until LEDs responded correctly to each lighting condition.
- This experiment demonstrated how physical changes (light) can be translated into electrical signals that a microcontroller can process.
- LDR-based sensing is commonly used in IoT applications such as automatic streetlights, smart home lighting, energy-saving systems, and environmental monitoring.